

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Regular / Supplementary Examinations Jan / Feb-2025

Common to EC / EE / EI / ET

LINEAR ALGEBRA, FOURIER TRANSFORMS AND STATISTICS

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of scientific calculator and hand book for mathematics is permitted.

PART-A

M CO BT

1	1.1	Let V be a vector space of all 2×2 matrices over $\mathbb{R}$. Suppose W consists of all matrices of V which have non-zero determinant, is W a subspace of V ? Give reason.	02	1	2
	1.2	Check whether the vectors $\{(-4, -3, -2), (-1, 2, 1), (2, 3, 4)\}$ are linearly independent or dependent.	02	3	4
	1.3	Find the image of the vector $(-3, 4)$, when it is rotated through an angle of 45 degree.	02	3	2
	1.4	In the SVD of the matrix $A = [-1, 2, 2]'$, $\Sigma =$ _____.	02	4	3
	1.5	The Fourier series coefficient a_0 of $f(x) = 2x^5/5$ in $(0, 2)$ is _____.	02	4	2
	1.6	The Fourier series of the function $f(x) = \begin{cases} 2+x, & 0 < x < 1 \\ 2-x, & 1 < x < 2 \end{cases}$ at the point of discontinuity, converges to _____.	02	4	1
	1.7	If the Fourier transform of e^{-x^2} is $\sqrt{\pi} e^{-\frac{s^2}{4}}$, then the Fourier transform of e^{-4x^2} is _____.	02	2	3
	1.8	The Fourier cosine transform of e^{-2x} is _____.	02	2	1
	1.9	If the first moment about an arbitrary value 2 in a distribution is 1, then the mean $\bar{x} =$ _____.	02	1	1
	1.10	If two regression equations of the variables x and y are $x = 19.13 - 0.87y$, $y = 11.6 - 0.5x$, then the mean values of x and y are _____ and _____.	02	1	2

PART-B

2	a	Find a bases for the four fundamental subspaces of the matrix $\begin{bmatrix} 2 & -4 & 1 & 2 & -2 & -3 \\ -1 & 2 & 0 & 0 & 1 & -1 \\ 10 & -4 & -2 & 4 & -2 & 4 \end{bmatrix}$.	08	1	1
	b	Obtain the range space, null space, rank and nullity of the linear transformation and verify the rank-nullity theorem, given $T: V_3(\mathbb{R}) \rightarrow V_4(\mathbb{R})$, defined by $T(e_1) = (0, 1, 0, 2)$, $T(e_2) = (0, 1, 1, 0)$, $T(e_3) = (0, 1, -1, 4)$.	08	2	2
3	a	Find a least-squares solution of the inconsistent system $Ax = b$ for $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$.	06	1	2

4	b	Obtain Singular Value Decomposition of the matrix $A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$.	10	4	3																											
	OR																															
	a	Find the QR factorization of the matrix $A = \begin{bmatrix} 1 & 3 & 5 \\ -1 & -3 & 1 \\ 0 & 2 & 3 \\ 1 & 5 & 2 \\ 1 & 5 & 8 \end{bmatrix}$.	08	1	2																											
	b	Diagonalize the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ by finding a non-singular matrix P and a diagonal matrix D such that $P^{-1}AP = D$.	08	4	3																											
5	a	Obtain the complex Fourier series for the function $f(x) = \begin{cases} 0, & \text{for } -\pi < x \leq 0 \\ \sin x, & \text{for } 0 < x < \pi \end{cases}$. Also sketch the graph of the function periodically.	08	3	2																											
	b	Analyze harmonically the data given below and express y in Fourier series up to the second harmonic:	08	4	4																											
		<table><tr><td>x</td><td>0</td><td>$\pi/3$</td><td>$2\pi/3$</td><td>π</td><td>$4\pi/3$</td><td>$5\pi/3$</td><td>2π</td></tr><tr><td>y</td><td>1.0</td><td>1.4</td><td>1.9</td><td>1.7</td><td>1.5</td><td>1.2</td><td>1.0</td></tr></table>	x	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π	y	1.0	1.4	1.9	1.7	1.5	1.2	1.0														
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6	a	Obtain the Fourier expansion of $f(x) = x^2$ over the interval $(-\pi, \pi)$. Deduce that $\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \infty$	08	3	2																											
	b	Expand $f(x) = x(2 - x)$ as half-range sine and cosine series over the interval $(0, 2)$.	08	4	4																											
7	a	Obtain the Fourier transform of the function $f(x) = \begin{cases} 1, & x \leq a \\ 0, & x > a \end{cases}$ where a is a positive constant. Evaluate $\int_0^\infty \frac{\sin \alpha}{\alpha} d\alpha$.	08	3	4																											
	b	Find the Fourier sine transform of $f(x) = e^{- x }$. Hence show that $\int_0^\infty \frac{x \sin mx}{1+x^2} dx = \frac{\pi}{2} e^{-m}, m > 0$.	08	2	3																											
		OR																														
8	a	Obtain the Fourier cosine and sine transform of $f(x) = e^{-ax}$, a is constant. Hence evaluate $\int_0^\infty \frac{t^2}{(t^2+1)^2} dt$ using Parseval's identity.	08	3	4																											
	b	Find the Fourier transform of the function $f(x) = e^{-x^2}$. Hence determine the Fourier transform of $e^{-x^2} \cos(2x)$ and $e^{-x^2} \sin(3x)$.	08	1	3																											
9	a	Calculate rank correlation coefficient of x and y from the following data:	08	3	2																											
		<table><tr><td>x</td><td>81</td><td>78</td><td>73</td><td>73</td><td>69</td><td>68</td><td>62</td><td>58</td></tr><tr><td>y</td><td>10</td><td>12</td><td>18</td><td>18</td><td>18</td><td>22</td><td>20</td><td>24</td></tr></table>	x	81	78	73	73	69	68	62	58	y	10	12	18	18	18	22	20	24												
	x	81	78	73	73	69	68	62	58																							
y	10	12	18	18	18	22	20	24																								
b	A set of experimental runs was made to determine a way of predicting cooking time y at various values of oven width x_1 and flue temperature x_2 . The coded data were recorded as follows:																															
		<table><tr><td>y</td><td>6.40</td><td>15.05</td><td>18.75</td><td>30.25</td><td>44.85</td><td>48.94</td><td>51.55</td><td>61.50</td></tr><tr><td>x_1</td><td>1.32</td><td>2.69</td><td>3.56</td><td>4.41</td><td>5.35</td><td>6.20</td><td>7.12</td><td>8.87</td></tr><tr><td>x_2</td><td>1.15</td><td>3.40</td><td>4.10</td><td>8.75</td><td>14.82</td><td>15.15</td><td>15.32</td><td>18.18</td></tr></table>	y	6.40	15.05	18.75	30.25	44.85	48.94	51.55	61.50	x_1	1.32	2.69	3.56	4.41	5.35	6.20	7.12	8.87	x_2	1.15	3.40	4.10	8.75	14.82	15.15	15.32	18.18			
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x_2	1.15	3.40	4.10	8.75	14.82	15.15	15.32	18.18																								
		Estimate the multiple linear regression equation $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$	08	2	3																											

OR

10 a

The wages of workers of a factory are as follows:

<i>Pay range (in thousand rupees)</i>	10-12	12-14	14-16	16-18	18-20	20-22	22-24
<i>No. of workers</i>	1	3	7	12	12	4	3

Calculate the first four central moments of the above distribution. Also compute β_1 and β_2

b

The following data represents carbon dioxide emission from coal-fluid boilers (in units of 1000 tons) over a period of years 1965 to 1977. Standardizing the year variable (x), the following table is obtained:

<i>Year (x)</i>	0	5	8	9	10	11	12
<i>CO₂ emission (y)</i>	910	680	520	450	370	350	340

Find the regression model of y on x and x on y .

08

3

2

08

2

3